

Class 13

Education and Human Capital

Part III — Mobilizing People and Expanding Opportunity

The American Economy — draft course materials (proposal under discussion)

The point of today's class

Education is one of America's most important opportunity-expanding institutions: it raises human capital, expands workers' outside options, and helps people stay ahead in the race with technology — but the measured premium is not automatically causal, and returns vary by school, field, cost, and completion.

Why it matters

- ▶ You are sitting inside the topic: college is a six-figure investment of tuition *and* foregone wages. Is it worth it?
- ▶ Human capital is productive capacity carried in people: literacy, coding, nursing, management, health, habits, credentials.
- ▶ American institutions did not only protect property and build markets — they **built people**. Mass education was one of America's distinctive growth institutions.
- ▶ Today is deliberately not “college is good”: it is about what education actually does — and how we know.

Today

1. Rung 3 of the opportunity ladder: learn
2. The fact: education pays — but is it causal?
3. Causal evidence: nudges into school
4. The high-school movement and the race between education and technology
5. Land-grant universities, Utah, and the GI Bill
6. Education after slavery — and the economics of access
7. Is college still worth it?

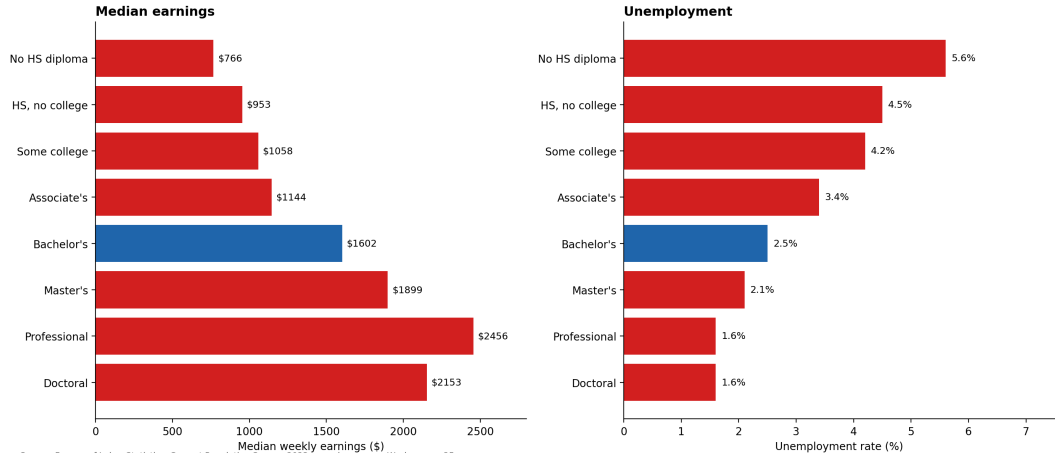
The opportunity ladder: rung 3 — learn

Rung		Class
8	rise	the American Dream and mobility
7	move	cities, housing, and place
6	belong	race, migration, and opportunity
5	choose	women, work, and family
4	arrive	immigration and assimilation
3	learn	education and human capital — today
2	bargain	labor markets and worker power
1	own your labor	free and coerced labor

- ▶ Class 12: bargaining power depends on outside options. Education is the **outside-option machine**: more occupations, more mobility, more ability to use technology, more resilience to shocks, less dependence on any one employer.

Education pays: earnings and unemployment by education

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Source: Bureau of Labor Statistics, "Education Pays" (2024).

Is that chart causal? Three theories of why education pays

- ▶ The fact (BLS 2024): earnings rise and unemployment falls with every education level — less than high school means median weekly earnings of \$738 and 6.2% unemployment.
- ▶ Strong motivation — but is it *proof* that education causes higher wages?

Theory	Claim	Example
Human capital	School teaches productive skills	Nurse, engineer
Signaling	School certifies ability, persistence	Diploma as screen
Networks / matching	School connects people to jobs	Internships, alumni

- ▶ All three can be true at once; the economic question is how much of the premium is productivity, signaling, or matching.

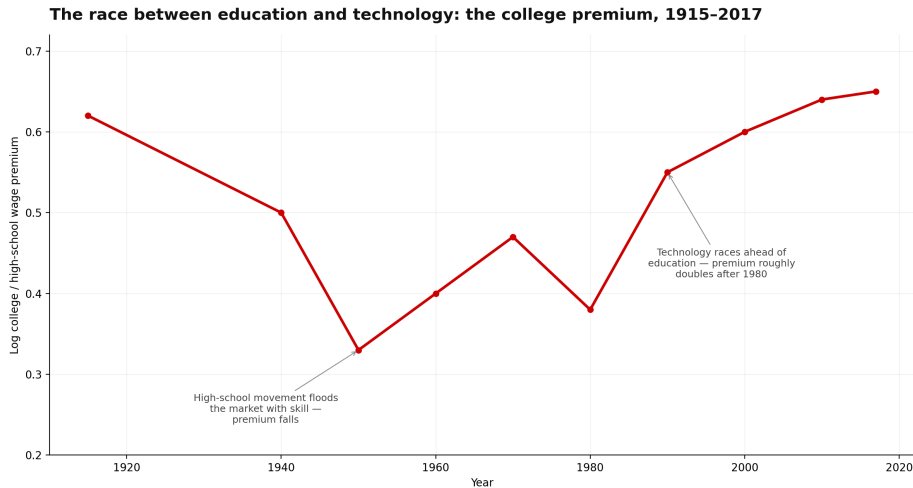
The selection problem — and the causal evidence

- ▶ If student A goes to college and out-earns student B, did college cause the gap — or were A and B different beforehand? **Selection**: the course's recurring problem.
- ▶ The trick: find people *nudged* into more schooling for reasons unrelated to ability.
 - ▶ Lleras-Muney: compulsory-attendance and child-labor laws, 1915–1939 — one more required year raised attainment by about 5% (similar for white males and females; not for Black students).
 - ▶ Angrist–Krueger: quarter of birth nudged some students into slightly more schooling; the estimated return was close to the simple comparison.
- ▶ Bottom line (Card; Oreopoulos–Petronijevic): education usually pays — for the average *and* the marginal student — but the size of the payoff varies.

The high-school movement and the race

- ▶ Goldin: Americans “invented the publicly funded, academic secondary school for the masses.” America’s human-capital edge was broad-based schooling, not just elite universities.
- ▶ Goldin–Katz frame inequality as a race:
 - ▶ technology raises the *demand* for skilled workers
 - ▶ education raises the *supply*
- ▶ When education outruns technology, the skill premium falls (the high-school era). When education slows and technology keeps going, the premium rises (after 1980).
- ▶ Key line: technology does not automatically make people richer — *people need skills to use it.*

The race between education and technology: the college premium, 1915–2017



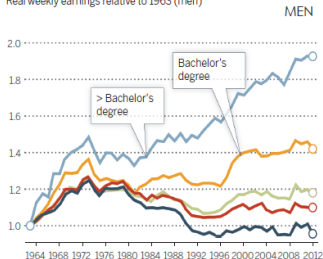
Reconstructed (approximate decade values) from Goldin and Katz (2008), *The Race between Education and Technology*, and Autor, Goldin, and Katz (2020). Teaching reconstruction — see originals for exact series.

The race between education and technology

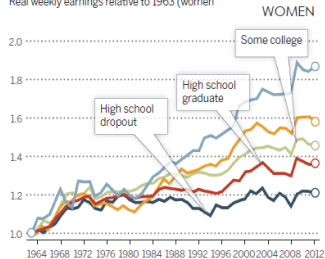
Indexed Real Full-Time Wages in U.S. by Sex and Education, 1963-2012

Changes in real wage levels of full-time U.S. workers by sex and education, 1963-2012

Real weekly earnings relative to 1963 (men)



Real weekly earnings relative to 1963 (women)



Autor, 2014

Source: Goldin and Katz, *The Race between Education and Technology*.

Public higher education: land grants, Utah, and the GI Bill

- ▶ Morrill Act (1862): federal land grants funded public colleges in “agriculture and the mechanic arts” — an American institution for *practical* human capital. (Honest complication: the granted lands were expropriated from Native communities.)
- ▶ Utah-local: the University of Utah traces to the University of Deseret (1850, first classes in private homes); Utah State (1888) is the state’s land-grant school. Even on the frontier, education was institution-building.
- ▶ GI Bill (1944): college funds, unemployment insurance, and housing for WWII veterans — higher education within reach of millions, though segregation and local administration meant benefits were not equally available in practice.

Education after slavery — and the economics of access

- ▶ Class 11 echo: freedpeople demanded schools immediately — Georgia's freedpeople's schools in early 1865 were described as “crowded to overflowing,” after laws had made it illegal to teach enslaved workers to read.
- ▶ Education is not merely a government service; it is an instrument of freedom and self-ownership.
- ▶ Hsieh–Hurst–Jones–Klenow again: improved allocation of talent among women and Black Americans explains 20–40% of growth in output per person, 1960–2010.
- ▶ Blocking people from education wastes talent twice: the economy loses output, and people lose lives they could have lived.

Is college still worth it?

Benefit	Cost
Higher expected earnings	Tuition
Lower unemployment risk	Foregone wages
Better job options and networks	Debt
Graduate and professional paths	Risk of noncompletion
Civic and personal benefits	Varies by major and program

- ▶ Oreopoulos–Petronijevic: college pays on average and for marginal students — but costs and benefits vary across students, schools, fields, and completion odds. It is an *investment under uncertainty*, not a guarantee.
- ▶ A recruiting note: economics is portable human capital — causal reasoning, data analysis, incentives, policy, markets.

Discussion

Four students: (A) full scholarship to an engineering program; (B) heavy borrowing for a program few students finish; (C) a community-college transfer path; (D) a paid apprenticeship. For each: what are the costs, the benefits, the opportunity cost, and the risk? What information would you want before choosing — and which part of the return is private vs. social?

Read before next class

- ▶ Goldin and Katz, *The Race between Education and Technology* (excerpt) — the framework behind today's killer chart.
- ▶ BLS, "Education Pays" — come ready to argue about whether the bars are causal.